# Matthew Knowles

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## Education

**University of Sheffield** Sheffield, UK

*MSc, Statistics with Medical Applications, Grade: Merit* *September 2022 – September 2024*

- Modules in time series, bayesian statistics, medical statistics, machine learning, and economic modelling
- Disseration: A Network Meta-Analysis of Treatments of Locally Advanced/Metastatic Pancreatic Cancer

**University of York** York, UK

*MMath (Hons), Mathematics, Grade: 2:1* *October 2018 – June 2022*

- Key modules included statistical pattern recognition, applied data science in R, and bayesian statistics
- Dissertation: Statistical Methods for Resetting Score Targets in Limited-Overs Cricket
- Received full colours for representing the university at cricket during my undergraduate studies

## Experience

**Biostatistics Manager** April 2026 – Present

*Astellas Pharma* *York, UK*

- **Statistician in the medcial affairs statistics group**
- Co-lead of a research group on indirect treatment-comparisons
- Opened a new internal service for indirect treatment comparison queries
- Participated on a collaborative research project with Yale University
- **Study-level statistician for women's health compound**
- Designed and generated TLF specifications for a phase IV longitudinal study
- Planned, conducted and delivered an indirect treatment-comparison using population adjustment methods
- **Study-level statistician for a novel oncology compound**
- Contributed to integrated evidence generation planning

**Associate Biostatistics Manager** December 2024 – March 2026

*Astellas Pharma* *York, UK*

- Spent ten months in clinical development, working on phase 1 trials for two novel oncology compounds
- Spent six months in the medical affairs statistics group, working on a women's health compound

**Senior Statistical Analyst** April 2024 – November 2024

*OPEN Health Evidence and Access* *London, UK*

- Lead the survival analysis workstream for a large clinical trial in chronic-lympochytic leukemia
- Supported the delivery of a network meta-analysis in crohn's disease for a NICE submission
- Conducted a non-proportional-hazards network meta analysis in ulcerative colitis
- Lead an indirect treatment comparison in crohn's disease for the Australian HTA body
- Implemented a novel survival modelling technique into a health-economic model
- Lead the development of in-house R packages for survival analysis and network meta-analysis

**Statistical Analyst** June 2022 – March 2024

*OPEN Health Evidence and Access* *York, UK*

- Worked on building and maintaining in-house R packages for survival analysis
- Performed and reported on the survival analyses of four clinical trials for submissions to health technology assessment agencies
- Performed a network meta-analysis in multiple myeloma
- Wrote statistical analysis plans for trial analyses and network meta-analyses

## Publications

**MSR48 Discrete-Event Simulation Model Projects Economic Outcomes Associated with Long-Term Clinical Remission in Patients with Severe Asthma Receiving Mepolizumab**, Value in Health, Volume 28, Issue 6, S283 (July 2025)

**Discrete Event-Simulation Model Projects Mepolizumab Treatment Outcomes Associated With Long-Term Clinical Remission in Severe Asthma**, American Journal of Respiratory and Critical Care Medicine, Issue 211 (May 2025)

**MSR213 Comparative Efficacy of Treatments in Advanced/Metastatic Pancreatic Cancer: Demonstrating the Power of Multi-Level Network Meta Regression**, Value in Health, Volume 27, Issue 12, S2 (December 2024)

## Skills

**Programming Languages**: R, Python, C/C++, Stan, SAS
**Developer Tools**: Git, Vim, VS Code, Rstudio Server, Rstudio Desktop
**Communicative Languages**: English, Norwegian
**Presenting**: Experienced at creating and giving talks/presentations in both Beamer and PowerPoint
**Other**: Excel, VBA, PowerBI, RShiny, R package development, LaTeX

## Professional Memberships

Royal Statistical Society

## Research Interests

Medical Statistics
- Application of ITCs to oncology, endocrinology, and emergency medicine
- Model diagnostics for ITC methods
- Population adjustment
- Estimands and target trial emulation

Therapeutic areas
- Pancreatic cancer
- Endocrinology

Computational Statistics